

Leaving Certificate Examination 2024

Mathematics

Higher Level

Paper 2

Marking scheme

300 marks

Marking Scheme – Paper 2, Section A and Section B

Structure of the marking scheme

Candidate responses are marked according to different scales, depending on the types of response anticipated. Scales labelled A divide candidate responses into two categories (correct and incorrect).

Scales labelled B divide responses into three categories (correct, partially correct, and incorrect), and so on. The scales and the marks that they generate are summarised in this table:

Scale label	A	B	C	D
No of categories	2	3	4	5
5 mark scales	0,5	0, 2, 5	0, 2, 3, 5	0, 2, 3, 4, 5
10 mark scales			0, 4, 6, 10	0, 3, 5, 7, 10
15 mark scale			0, 6, 8, 15	0, 4, 6, 8, 15
20 mark scale				0, 8, 11, 13, 20

A general descriptor of each point on each scale is given below. More specific directions in relation to interpreting the scales in the context of each question are given in the scheme, where necessary.

Marking scales – level descriptors

A-scales (two categories)

- incorrect response
- correct response

B-scales (three categories)

- response of no substantial merit
- partially correct response
- correct response

C-scales (four categories)

- response of no substantial merit
- response with some merit
- almost correct response
- correct response

D-scales (five categories)

- response of no substantial merit
- response with some merit
- response about half-right
- almost correct response
- correct response

Palette of annotations available to examiners

Symbol	Name	Meaning in the body of the work	Meaning when used in the right margin
	Tick	Work of relevance	The work presented in the body of the script merits full credit
	Cross	Incorrect work (distinct from an error)	The work presented in the body of the script merits 0 credit
	Star	Rounding / Unit / Arithmetic error / Misreading	
	Horizontal wavy	Error	
	P		The work presented in the body of the script merits <i>Partial Credit</i>
	L		The work presented in the body of the script merits <i>Low Partial Credit</i>
	M		The work presented in the body of the script merits <i>Mid Partial Credit</i>
	H		The work presented in the body of the script merits <i>High Partial Credit</i>
	F star		The work presented in the body of the script merits <i>Full Credit – 1</i>
	Left Bracket		Another version of this solution is presented elsewhere and it merits equal or higher credit
	Vertical wavy	No work on this page / portion of this page	
	Oversimplify	The candidate has oversimplified the work	
	Work of merit	The candidate has produced work of merit (in line with that defined in the scheme)	
	Stops early	The candidate has stopped early in this part	

Note: Where work of substance is presented in the body of the script, the annotation on the right margin should reflect a combination of annotations in the work.

In a **C scale** that is **not** marked using steps, where * and and appear in the body of the work, then should be placed in the right margin.

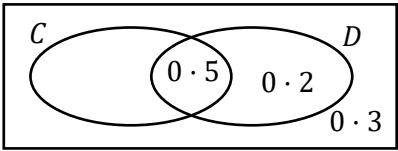
In the case of a **D scale** with the same annotations, should be placed in the right margin.

Detailed marking notes

Model Solutions & Marking Notes

Note: The model solutions for each question are not intended to be exhaustive – there may be other correct solutions. Any Examiner unsure of the validity of the approach adopted by a particular candidate to a particular question should contact his / her Advising Examiner.

Q1	Model Solution – 30 Marks	Marking Notes
(a)	(i) Mode = 34 so $a = 4$ (ii) Range = 49 ... 69 – 20 = 49 So $b = 0$ and $c = 9$ (iii) Median = $\frac{"4d" + 45}{2} = 43.5$ $"4d" + 45 = 87$ $"4d" = 42$ So $d = 2$	Scale 15D (0, 4, 6, 8, 15) Note: solution requires 4 values: a, b, c, and d Full credit for correct answers without work. <i>Low Partial Credit:</i> • Work of merit in finding one value, for example, indicates median is between "4d" and 45 <i>Mid Partial Credit</i> • Two values correct • One value correct and work of merit in finding two other values <i>High Partial Credit:</i> • Three values correct • Two values correct and work of merit in finding the other two values <i>Full Credit –1</i> • Apply a * if both digits are given instead of just second digit in parts (ii) and (iii), once only. Do not penalise thereafter
(b) (c)	(b) A valid description, for example: Everyone improved but the stronger swimmers improved to a greater extent than the weaker swimmers. (c) $r = 0.91451 \dots = 0.9145$ [4 D.P.]	Scale 15C (0, 6, 8, 15) In (b), needs to mention both increase for these students and non-linearity of increase (for example, "the better swimmers improved more") to be considered correct. <i>Low Partial Credit:</i> • Work of merit in (b), for example, mentions that swimmers improved, or reference to non-linearity <i>High Partial Credit</i> • (b) or (c) correct <i>Full Credit –1</i> • Apply a * for incorrect rounding in (c) (incorrect to 4 DP, or correct to a different number of DP)

Q2	Model Solution – 30 Marks	Marking Notes
(a)	$(0)(0.3) + (2)(0.4) + (x - 10)(0.28) + (x)(0.02) = 10$ $0.8 + 0.28x - 2.8 + 0.02x = 10$ $0.3x = 12$ $x = 40$	Scale 10C (0, 4, 6, 10) <i>Low Partial Credit:</i> - Work of merit, for example, some correct term in $E(x)$ <i>High Partial Credit:</i> - Fully correct equation
(b)	$P(A \cap B) = 0$ $P(A \cup B) = 0.5$	Scale 5B (0, 2, 5) <i>Partial Credit:</i> Work of merit, for example, correct relevant formula, such as $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; draws Venn diagram with some relevant use of 0.1 and/or 0.4
(c)	To maximise $(C \cup D)'$ need to max $(C \cap D)$  Max $P[(C \cup D)'] = 0.3$	Scale 10C (0, 4, 6, 10) <i>Low Partial Credit:</i> - Work of merit, for example, Venn diagram with some relevant use of 0.5 and/or 0.7 - States that $C \cap D$ needs to be maximised or that $C \cup D$ needs to be minimised - Indicates that one set is a subset of the other <i>High Partial Credit:</i> $C \cap D = 0.5$
(d)	If it is raining I am more likely to wear a coat than if it isn't	Scale 5A (0, 5)

Q3	Model Solution – 30 Marks	Marking Notes
(a)	Area of $ABC = \frac{1}{2}(10)(13) \sin 110 = 61.08..$ Area of $ABCD = 2(61.08 ...)$ $= 122.16 ...$ $= 122 \text{ [cm}^2\text{]} \text{ [nearest cm}^2\text{]}$ OR Area of $ABCD = (10)(13) \sin 110$ $= 122.16 ...$ $= 122 \text{ [cm}^2\text{]} \text{ [nearest cm}^2\text{]}$	Scale 15C (0, 6, 8, 15) <i>Low Partial Credit:</i> • Work of merit, for example, triangle formula with some correct substitution; formula ah with some correct substitution or further work; formula $ab \sin C$ with some correct substitution; a correctly labelled diagram <i>High Partial Credit:</i> • Fully correct substitution into parallelogram formula • Fully correct substitution in area of triangle formula <i>Full Credit-1:</i> • No rounding or incorrect rounding • Calculator in incorrect mode
(b)	Reference angle in 1st and 4th quadrants $\cos^{-1} \frac{\sqrt{3}}{2} = 30^\circ \text{ or } 330^\circ$ $2X = 30^\circ, 330^\circ, 390^\circ, 690^\circ$ $X = 15^\circ, 165^\circ, 195^\circ, 345^\circ$	Scale 5D (0, 2, 3, 4, 5) <i>Low Partial Credit:</i> • Work of merit in finding reference angle <i>Mid Partial Credit:</i> • One correct value of X found • Two correct values of $2X$ found <i>High Partial Credit:</i> • Two correct values of X found • All 4 correct values of $2X$ found but fails to find X
(c)	$\frac{\sin \theta}{45} = \frac{\sin 25}{15\sqrt{3}}$ $\sin \theta = \frac{45(0.4226)}{15\sqrt{3}} = 0.73196$ $\sin \theta = 0.73196 ... 47.05^\circ$ angle in 1st and 2nd quadrants $\theta = 47.05^\circ \text{ or } \theta = 132.95^\circ$ $\theta = 47^\circ \text{ or } \theta = 133^\circ \text{ [nearest degree]}$	Scale 10D (0, 3, 5, 7, 10) Accept correct answer without units (degrees) <i>Low Partial Credit:</i> • Work of merit, for example, some correct substitution into the Sine rule, or correctly labelled diagram <i>Mid Partial Credit:</i> • Sine rule with fully correct substitution <i>High Partial Credit:</i> • One correct angle found <i>Full Credit-1</i> • Apply a * if not rounded correctly

Q4	Model Solution – 30 Marks	Marking Notes
(a)	<i>Correct construction</i>	Scale 15D (0, 4, 6, 8, 15) <i>Low Partial Credit:</i> • Work of merit in finding the midpoint of any side, for example, measures the length of one side <i>Mid Partial Credit:</i> • Midpoints of 2 sides identified (construction/measurements shown) • One median drawn with no construction/measurements shown <i>High Partial Credit:</i> • One median drawn (construction/measurements shown) • 2 medians with no construction
(b)	<i>Any valid proof, for example:</i> Construction: Draw a line through B parallel to DF. Where this cuts AD and CF, label X and Y, respectively. Proof: $AB = BC$... given $\angle XAB = \angle BCY$... alternate $\angle XBA = \angle CBY$... vertically opposite So $\triangle XBA \equiv \triangle YBC$... ASA Therefore $BX = BY$... congruency But $DEBX$ is a parallelogram, as the opposite sides are parallel. So $BX = DE$... opposite sides of a parallelogram Similarly, $BY = EF$. So $DE = EF$.	Scale 15D (0, 4, 6, 8, 15) <i>Low Partial Credit:</i> • Work of merit, for example, correct construction indicated, identifies parallelograms, reference to congruency <i>Mid Partial Credit:</i> • Construction, and 2 correct steps in establishing congruency of $\triangle XBA$ and $\triangle YBC$ (no justifications) <i>High Partial Credit:</i> • Construction, and 3 correct steps (including correct relevant side) in establishing congruency of $\triangle XBA$ and $\triangle YBC$, with at least one justification • Fully correct logical statements, but no justifications

Q5	Model Solution – 30 Marks	Marking Notes
(a) (i)	$x^2 + y^2 + 4x - 6y + 5 = 0$ $g = 2, \quad f = -3$ Centre = $(-2, 3)$ $r = \sqrt{g^2 + f^2 - c} = \sqrt{(2)^2 + (-3)^2 - 5}$ $= 2\sqrt{2}$	Scale 10D (0, 3, 5, 7, 10) <i>Low Partial Credit:</i> • Work of merit in finding centre or radius, for example, identifies g, f or c <i>Mid Partial Credit</i> • Centre correct • Radius correct <i>High Partial Credit:</i> • One of centre or radius correct and work of merit in finding the other
(a) (ii)	Centre of $c = (2, -1)$ Radius of $c = \sqrt{72} = 6\sqrt{2}$ Distance between centres $= \sqrt{(2 - (-2))^2 + (-1 - 3)^2}$ $= \sqrt{16 + 16}$ $= 4\sqrt{2}$ Difference between radii $= 6\sqrt{2} - 2\sqrt{2}$ $= 4\sqrt{2}$ $4\sqrt{2} = 4\sqrt{2}$ Therefore, circles touch internally	Scale 10D (0, 3, 5, 7, 10) Consider the solution as having 4 steps before the conclusion: 1. Find centre of circle c 2. Find radius of circle c 3. Find distance between the centres of circle c and circle s 4. Find difference in radii <i>Low Partial Credit:</i> • Work of merit in any one of the 4 steps <i>Mid Partial Credit</i> • 2 correct steps <i>High Partial Credit:</i> • 3 correct steps <i>Full Credit –1</i> • Apply a * if there is no conclusion or an incorrect conclusion

Q5	Model Solution – 30 Marks	Marking Notes
(b)	Centre = $(9, k)$ So distance from $(7, 10)$ to $(9, k)$ = distance from $(12, 8)$ to $(9, k)$ $\sqrt{(9-7)^2 + (k-10)^2} = \sqrt{(12-9)^2 + (8-k)^2}$ $4 + k^2 - 20k + 100 = 9 + 64 - 16k + k^2$ $4k = 31$ $k = \frac{31}{4}$ So, centre = $(9, \frac{31}{4})$ $\text{Radius} = \sqrt{(9-7)^2 + \left(\frac{31}{4} - 10\right)^2} = \frac{\sqrt{145}}{4}$ $\text{Eqn: } (x-9)^2 + \left(y - \frac{31}{4}\right)^2 = \frac{145}{16}$ OR Centre = $(-g, -f) = (9, k) \therefore g = -9 \dots \text{Eqn A}$ $(7, 10): 7^2 + 10^2 + 2g(7) + 2f(10) + c = 0 \dots \text{Eqn B}$ So: $20f + c = -23$ $(12, 8): 12^2 + 8^2 + 2g(12) + 2f(8) + c = 0 \dots \text{Eqn C}$ So: $16f + c = 8$ From B and C: $f = \frac{-31}{4}, \quad c = 132$ Eqn: $x^2 + y^2 - 18x - \frac{31}{2}y + 132 = 0$ OR Method 3 Consider the chord joining $(7, 10)$ and $(12, 8)$ Midpoint = $(\frac{19}{2}, 9)$, Slope = $-\frac{2}{5}$ Therefore Eq. of line perpendicular to chord passing through the point $(\frac{19}{2}, 9) : y - 9 = \frac{5}{2}(x - \frac{19}{2})$ $(9, k) \in y - 9 = \frac{5}{2}(x - \frac{19}{2})$ $k - 9 = \frac{5}{2}(9 - \frac{19}{2})$ $k = \frac{31}{4}$ So, centre = $(9, \frac{31}{4})$ $\text{Radius} = \sqrt{(9-7)^2 + \left(\frac{31}{4} - 10\right)^2} = \frac{\sqrt{145}}{4}$ $\text{Eqn: } (x-9)^2 + \left(y - \frac{31}{4}\right)^2 = \frac{145}{16}$	Scale 10D (0, 3, 5, 7, 10) <i>Low Partial Credit:</i> • Work of merit in finding the centre/ radius • Some substitution into general equation of a circle • Work of merit in finding midpoint or slope of chord joining $(7, 10)$ to $(12, 8)$ <i>Mid Partial Credit</i> • $\sqrt{(9-7)^2 + (k-10)^2}$ and $\sqrt{(12-9)^2 + (8-k)^2}$ • Two of <i>Eqns A, B, or C</i> • Finds equation of the line perpendicular to the chord joining $(7, 10)$ to $(12, 8)$ <i>High Partial Credit:</i> • Finds $k = \frac{31}{4}$ (Method 1 or 3) • Two independent equations in f and c (Method 2)

Q6	Model Solution – 30 Marks	Marking Notes
(a)	$(1, 13) \rightarrow +5$ and $-2 \rightarrow (6, 11)$ $(6, 11) \rightarrow +3(5)$ and $3(-2) \rightarrow B(21, 5)$ OR $(1, 13) \rightarrow +5$ and $-2 \rightarrow (6, 11)$ $(1, 13) \rightarrow +4(5)$ and $4(-2) \rightarrow B(21, 5)$ OR $B = (x_2, y_2), m:n = 1:3$ $(6, 11) = \left(\frac{1(x_2)+3(1)}{1+3}, \frac{1(y_2)+3(13)}{1+3} \right)$ $6 = \frac{x_2+3}{4}$ and $11 = \frac{y_2+39}{4}$ $x_2 = 21$ and $y_2 = 5$ $B = (21, 5)$	Scale 5D (0, 2, 3, 4, 5) <i>Low Partial Credit:</i> • Work of merit, for example, identifies one correct part of translation; some correct substitution into formula; relevant diagram with labelling <i>Mid Partial Credit:</i> • Initial translation correct • Fully correct substitution into formula • Treats C as midpoint, otherwise correct. <i>High Partial Credit:</i> • Identifies trebling/quadrupling of initial translation • $6 = \frac{x_2+3}{4}$ and $11 = \frac{y_2+39}{4}$, or equivalent • x_2 correct or y_2 correct
(b)	$\frac{4}{3}x - y - 11 = 0$ Perpendicular distance: $= \frac{\left \frac{4}{3}(5) + (-1)(-2) - 11 \right }{\sqrt{\left(\frac{4}{3}\right)^2 + (-1)^2}}$ $= \frac{ -7 }{5}$ $= 1.4$ [units]	Scale 10C (0, 4, 6, 10) Consider the solution as having 3 steps: 1. Rewrite the equation in the form $ax + by + c = 0$ 2. Substitute into formula 3. Evaluate <i>Low Partial Credit:</i> • Work of merit in one step, for example, some correct transposition <i>High Partial Credit:</i> • 2 steps correct <i>Full Credit –1:</i> • Apply a * if the modulus is omitted

Q6	Model Solution – 30 Marks	Marking Notes
(c)	(i) $\binom{16}{2} = 120$ [pairs] Also accept $16 \times 15 = 240$ [pairs], i.e. where a pair is considered as being ordered (ii) $\binom{4}{2} \times 4 = 24$ [unordered pairs that give horizontal lines] So probability = $\frac{24}{120}$ or $\frac{1}{5}$ OR $(4 \times 3) \times 4 = 48$ [ordered pairs that give horizontal lines] So probability = $\frac{48}{240}$ or $\frac{1}{5}$	Scale 15D (0, 4, 6, 8, 15) Accept solutions where pairs are considered unordered or ordered. Note that the answer in (c)(ii) should be the same regardless. If pairs are treated as ordered in one part and unordered in the other, award <i>HPC</i> at most. Note: Accept for full credit correct answers without work <i>Low Partial Credit:</i> • Work of merit, for example, in (i), uses 16 in calculation, or, in (ii), gets some number divided by 120 or 240, as appropriate; or draws horizontal lines joining given points on diagram <i>Mid Partial Credit</i> • One part correct • Work of merit in both parts. Note that WOM in the diagram can only count as WOM in one part <i>High Partial Credit</i> • One part correct and work of merit in the other part

Q7	Model Solution – 50 Marks	Marking Notes
(a) (i)	$z = \frac{50 - 48.2}{10.6} = 0.1698$ $P(z < 0.17) = 0.5675 \text{ or } 56.75\%$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> - Work of merit, for example, some correct substitution into relevant formula, relevant diagram drawn, indicates μ or σ <i>High Partial Credit:</i> - Finds z-score $\left(\frac{50-48.2}{10.6}\right)$
(a)(ii)	Oldest 10% $\therefore$ 90% less than 'A' years old So, $z = 1.28$ (or 1.29) $1.28 = \frac{A - 48.2}{10.6}$ $A = 61.768$ Using $z = 1.29, A = 61.874$ $A = 62$ [nearest whole number]	Scale (0, 4, 6, 10) Consider solution as requiring 3 steps: - Finds z-value - Writes equation in A - Finds value of A <i>Low Partial Credit:</i> - Work of merit, for example, mentions 90%, or finds z-value of 10% <i>High Partial Credit:</i> 2 correct steps <i>Full Credit –1</i> - Apply a * for incorrect or no rounding
(b) (i)	2 successes: $\binom{6}{2} \left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^4 = \frac{768}{3125} = 0.24576$	Scale 5C (0, 2, 3, 5) Note: No penalty applies if answer is rounded to either 3 or 4 decimal places <i>Low Partial Credit:</i> - WOM, for example, mentions $\frac{4}{5}$ or $\binom{6}{2}$; or $P(\text{success}) = \frac{1}{5}$ <i>High Partial Credit</i> $\binom{6}{2} \left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^4$ $\left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^4$ evaluated

Q7	Model Solution – 50 Marks	Marking Notes
(b)(ii)	$\binom{n}{0} \left(\frac{1}{5}\right)^0 \left(\frac{4}{5}\right)^n = 0.0047$ $0.8^n = 0.0047$ $n = \frac{\ln 0.0047}{\ln 0.8}$ $n = 24.021..$ $n = 24$	Scale 5C (0, 2, 3, 5) Accept $n = 24$ if it is indicated that $0.8^{24} = 0.0047 \dots$ for Full Credit <i>Low Partial Credit:</i> - Work of merit, for example, mentions $\binom{n}{0}$, $\left(\frac{1}{5}\right)^0$ or $\frac{4}{5}$ <i>High Partial Credit</i> - Fully correct equation <i>Full Credit -1</i> - Apply a * for answer not rounded correctly
(c)	Booked: $(0.45) \left(\frac{1}{3}\right) + (0.55) \left(\frac{2}{5}\right) = \frac{37}{100}$ Booked on new system: $(0.55) \left(\frac{2}{5}\right) = \frac{11}{50}$ Probability = $\frac{\left(\frac{11}{50}\right)}{\left(\frac{37}{100}\right)} = \frac{22}{37} = 59\% \text{ } [\in \mathbb{N}]$	Scale 15D (0, 4, 6, 8, 15) <i>Low Partial Credit:</i> - Work of merit, for example, one correct relevant multiplication - Tree diagram with some correct value(s) <i>Mid Partial Credit:</i> $\frac{37}{100}$ or $\frac{11}{50}$ (or equivalent) <i>High Partial Credit</i> $\frac{37}{100}$ and $\frac{11}{50}$ <i>Full Credit -1</i> - Apply a * for answer not in correct form

Q7	Model Solution – 50 Marks	Marking Notes
(d)	Null Hypothesis: $p = 0.75$ or similar (for example, “in 2024, PK Hotels were rated the best hotel chain in Europe by 75% of their customers”) Alternative Hypothesis: $P \neq 0.75$ or similar Calculations: Conf Int: $0.765 \pm \frac{1}{\sqrt{1000}} = 0.765 \pm 0.0316...$ So $0.7334 < p < 0.7966$ Conclusion: Fail to reject H_0 or there is not enough evidence to conclude that this percentage has changed. Accept: the percentage has not changed, or similar Reason: 0.75 is within the interval Note: accept the margin of error based on $1.96 \sqrt{\frac{(0.75)(0.25)}{1000}} = 0.0268... \text{ or }$ $1.96 \sqrt{\frac{(0.765)(0.235)}{1000}} = 0.0262...$	Scale 10D (0, 3, 5, 7, 10) Note: 4 steps to check: 1. Hypotheses 2. Calculations 3. Conclusion 4. Reason Reason may be included with Calculations (or Conclusion), and must be based on relevant calculations – if either a confidence interval or test statistic is not found, a conclusion cannot be drawn If the null or alternative hypothesis is incorrect, then Step 1 is not considered correct. In this case, mark Steps 2 and 4 as if the null and alternative hypothesis had been given. However, Step 3 (conclusion) cannot be considered correct, so award <i>Mid Partial Credit</i> at most. <i>Low Partial Credit:</i> • Work of merit in one step <i>Mid Partial Credit:</i> • Two steps correct <i>High Partial Credit</i> • Three steps correct

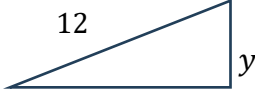
Q8	Model Solution – 50 Marks	Marking Notes
(a)	One dimension: 15 [cm] Other dimension: $2\pi r = 2\pi(5) = 31.41 \dots = 31.4$ [cm] [1 DP] Dimensions can be presented in either order.	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> • Work of merit in finding length, for example, circumference formula • Width correct • $2\pi r h = 150\pi$ <i>High Partial Credit</i> • Width correct and work of merit in finding length <i>Full Credit-1</i> • Apply a * if dimensions are not rounded correctly to one decimal place where appropriate
(b)	$r^2 = 6^2 + 11^2$ $r = \sqrt{36 + 121} = \sqrt{157}$ $\text{Vol} = \frac{4}{3}\pi(\sqrt{157})^3$ $= 8240.20 \dots$ $= 8240.2 \text{ [cm}^3\text{]} \text{ [1 DP]}$	Scale 5D (0, 2, 3, 4, 5) 4 steps: 1. Sets up Pythagoras 2. Finds r 3. Subs into volume formula 4. Evaluates volume <i>Low Partial Credit:</i> • Work of merit, for example, radius of cylinder = 6cm <i>Mid Partial Credit</i> • 2 steps correct • 1 step correct and work of merit in 2 other steps <i>High Partial Credit</i> • 3 steps correct

Q8	Model Solution – 50 Marks	Marking Notes
(c) (i) (ii)	(i) Any valid reason given, for example: Opposite angles in a cyclic quadrilateral sum to 180° OR Angle at the circumference standing on the diameter is 90° (ii) $\angle CEB = \angle DEB$... both 90° $\angle ECB = 90 - \angle EDB$ $\angle DBE = 90 - \angle EDB$ So $\angle ECB = \angle DBE$ So $\angle CBE = \angle EDB$... angles in Δ add to 180° OR $\angle CEB = \angle DEB$... both 90° $\angle CAB = \angle CBE$... isosceles ΔACB $\angle CAB = \angle CDB$... both standing on arc CB So $\angle CBE = \angle CDB$ So $\angle EBD = \angle ECB$... angles in Δ add to 180°	Scale 20D (0, 8, 11, 13, 20) Much of the work for (ii) may be presented on the diagram. <i>Low Partial Credit:</i> - Part (i) correct - Work of merit in part (ii), for example, identifies 2 equal angles <i>Mid Partial Credit</i> - Part (i) correct and one pair of corresponding angles in triangles BCE and DBE identified in Part (ii) - Two pairs of corresponding angles in triangles BCE and DBE identified in Part (ii) <i>High Partial Credit</i> (ii) correct (with justification for $\angle ECB = \angle DBE$ or $\angle CBE = \angle EDB$) (i) correct and two pairs of corresponding angles in triangles BCE and DBE identified in Part (ii)
(c)(iii)	$\frac{r}{20-h} = \frac{h}{r}$ $r^2 = h(20-h)$ $r^2 = 20h - h^2$	Scale 10C (0, 4, 6, 10) <i>Low Partial Credit:</i> - Work of merit in setting up ratio, for example $ED = 20 - h$ <i>High Partial Credit</i> $\frac{r}{20-h} = \frac{h}{r}$ <i>Full Credit –1</i> $r^2 = h(20-h)$
(c)(iv)	$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(20h - h^2)h$ $= \frac{1}{3}\pi(20h^2 - h^3)$ $\frac{dV}{dh} = \frac{1}{3}\pi(40h - 3h^2)$ $\frac{1}{3}\pi(40h - 3h^2) = 0 \text{ at local max/min}$ $40h - 3h^2 = 0$ $h(40 - 3h) = 0$	Scale 10C (0, 4, 6, 10) Consider solution involving 3 steps: 1: Express V in terms of π and h 2: Find $\frac{dV}{dh}$ 3: Solve for h, when $\frac{dV}{dh} = 0$ <i>Low Partial Credit:</i> - Work of merit, for example, some correct differentiation or substitution <i>High Partial Credit</i> 2 steps correct

Q8	Model Solution – 50 Marks	Marking Notes
	$h = 0, h = \frac{40}{3}$ Answer: $h = \frac{40}{3}$ [cm] [as $h = 0$ gives $V = 0$]	One step correct and work of merit in other 2 steps

Q9	Model Solution – 50 Marks	Marking Notes
(a) (i)	$(x - 1)^2 + (y - 17)^2 = 144$	Scale 5B (0, 2, 5) <i>Partial Credit:</i> Work of merit, for example, some correct substitution into equation of circle; or indication of radius on the diagram
(a)(ii)	$(a - 1)^2 + (8 - 17)^2 = 144$ $(a - 1)^2 = 63$ $a - 1 = \sqrt{63}$ (as $a > 0$) $= 1 + \sqrt{63}$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> - Work of merit, for example, some correct substitution into equation <i>High Partial Credit:</i> $a^2 - 2a - 62 = 0$ $(a - 1)^2 = 63$ <i>Full Credit-1</i> - Apply a * if $1 - \sqrt{63}$ is also presented as a solution
(a)(iii)	$ PC = \sqrt{(10 - 1)^2 + (6 - 17)^2} = \sqrt{202}$ $ PC - \text{radius} = \sqrt{202} - 12 = 2.2126 \dots$ Shortest distance = $2.2126 \dots \times 100$ $= 221.26 \dots$ $= 221 \text{ [m]}$ [nearest m]	Scale 10C (0, 4, 6, 10) <i>Low Partial Credit:</i> - Work of merit, for example, some correct substitution into distance/perpendicular distance formula; finds slope of CP; <i>High Partial Credit:</i> $PC - \text{radius} = \sqrt{202} - 12$ or equivalent <i>Full Credit –1</i> - Apply a * for no rounding or incorrect rounding
(b)	$x = 13$	Scale 5B (0, 2, 5) <i>Partial Credit:</i> Work of merit, for example, some indication that equation is of form $x = \text{constant}$, or draws in relevant tangent or normal

Q9	Model Solution – 50 Marks	Marking Notes
(c)	Step 1: Slope of $w = \frac{1}{3}$ Step 2: $\perp$ slope $= -3$ Step 3: $y - 6 = -3(x - 10)$ Step 4: $3x + y = 36$ $3x + y = 36 \quad (\times 3)$ $\underline{x - 3y = 9}$ $9x + 3y = 108$ $\underline{x - 3y = 9}$ $10x = 117$ $x = 11.7$ Step 5: $11.7 - 3y = 9$ $y = 0.9$ (11.7, 0.9)	Scale 10D (0, 3, 5, 7, 10) Consider the solution involving 5 steps: 1. Find the slope of w 2. Find the perpendicular slope 3. Find the equation of the line through P that is perpendicular to w 4. Find one co-ordinate of the point of intersection of the two lines 5. Find the other co-ordinate of the point of intersection of the two lines Note: $w: x - 3y = 9$, therefore eq. of $\perp$ line is $3x + y + c = 0$ (Step 1 & Step 2) <i>Low Partial Credit:</i> • Work of merit, for example, $-3y = -x + 9$ <i>Mid Partial Credit</i> • 2 steps correct and work of merit in another step <i>High Partial Credit</i> • 4 steps correct

Q9	Model Solution – 50 Marks	Marking Notes
(d)	$y^2 + (3y)^2 = (12)^2$ $10y^2 = 144$ $y = \sqrt{14.4} = 3.8 \dots$ $x = 9 + 3\sqrt{14.4} = 20.4 \dots$ $(20.4, 3.8)$ [1 DP] OR $x^2 = 3^2 + 1^2$ $x^2 = 10$ $x = \sqrt{10}$ $1200\text{m} = 12 \text{ units}$ Factor increase = $\frac{12}{\sqrt{10}} = 3.794\dots$ $x = 3 \times 3.794 + 9 = 20.38\dots$ $y = 3.794\dots$ $(20.4, 3.8)$ [1 DP] OR  Slope of line $w = \frac{1}{3}$ Angle between line w and the x – axis = $\tan^{-1} \frac{1}{3} = 18.435$ Then, $\sin 18.435 = \frac{y}{12}$ $y = 3.794 \dots$ $x = 3.794 \times 3 + 9 = 20.384 \dots$ $(20.4, 3.8)$ [1 DP]	Scale 15D (0, 4, 6, 8, 15) <i>Low Partial Credit:</i> • Work of merit in identifying inputs for Pythagoras; slope of $w = \frac{1}{3}$; $1200\text{m} = 12 \text{ units}$ <i>Mid Partial Credit:</i> • $y^2 + (3y)^2 = (12)^2$ or equivalent • Finds angle between line w and the positive direction of the x – axis <i>High Partial Credit</i> • Finds x or y value <i>Full Credit-1</i> • Apply a * for no rounding or incorrect rounding, once only

Q10	Model Solution – 50 Marks	Marking Notes
(a)(i)	$ OB = ON = 100 + 20 = 120$	Scale 5B (0, 2, 5) <i>Partial Credit:</i> Work of merit, for example, states $OB = ON$
(a)(ii)	$\cos \angle BOT = \frac{90}{120}$ $\angle BOT = \cos^{-1} \frac{90}{120} = 41.40 \dots^\circ = 41.4^\circ$ [1 DP]	Scale 10C (0, 4, 6, 10) <i>Low Partial Credit:</i> - Work of merit, for example, some correct substitution into a trigonometric ratio <i>High Partial Credit</i> $\cos \angle BOT = \frac{90}{120}$ or equivalent
(a)(iii)	$ \angle BOB' = 180 - 2(41.4) = 97.2^\circ$ $\sin 41.4 = \frac{20}{ OA }$ $ OA = \frac{20}{\sin 41.4} = 30.24 \dots$ $\text{Area } ABB'A' = \text{Area } BOB' - \text{Area } AOA'$ $= \frac{97.2}{360} \pi (120)^2 - \frac{97.2}{360} \pi (30.24)^2$ $= 11438.84 \dots = 11\,439 \text{ [cm}^2\text{]} \text{ [}\in \mathbb{N}\text{]}$	Scale 10D (0, 3, 5, 7, 10) Consider solution as involving 4 steps: 1: Finds $ OA $ 2: Finds $ \angle BOB' $ 3: Finds area of one sector (either BOB' or AOA') 4: Finds area of second sector and finishes <i>Low Partial Credit:</i> - Work of merit in one step, for example, finds $\angle AOM$ <i>Mid Partial Credit</i> - One step correct and work of merit in another step <i>High Partial Credit</i> - Three steps correct

Q10	Model Solution – 50 Marks	Marking Notes
(b)	Method 1: $180^2 = x^2 + x^2 - 2x^2 \cos 105$ $180^2 = x^2(2 - 2 \cos 105)$ $x = \frac{180}{\sqrt{2 - 2 \cos 105}} = 113.44 \dots$ $= 113.4 \text{ [cm] [1 DP]}$ OR Method 2: $ \angle OE'E = \frac{180 - 105}{2} = 37.5^\circ$ $x^2 = x^2 + 180^2 - 2(x)(180) \cos 37.5$ $2(x)(180) \cos 37.5 = 180^2$ $x = \frac{32400}{360 \cos 37.5} = 113.44 \dots$ $= 113.4 \text{ [cm] [1 DP]}$	Scale 10D (0, 3, 5, 7, 10) Consider solution as involving 3 steps: Method 1: 1. Substitutes into Cosine Rule 2. Isolates x^2 3. Evaluates x OR Method 2: 1. Finds angle 37.5° 2. Substitutes into Cosine Rule 3. Evaluates x Note: Award <i>Mid Partial Credit</i> at most if the Cosine Rule is not used to find x <i>Low Partial Credit:</i> • Work of merit, for example, some correct substitution into Cosine Rule, or correctly indicates x on the diagram <i>Mid Partial Credit</i> • 1 step correct <i>High Partial Credit</i> • 2 steps correct
(c)(i)	(i) $3^5 = 243$	Scale 5B (0, 2, 5) Accept correct answer without work • Work of merit, for example, lists some correct patterns

Q10	Model Solution – 50 Marks	Marking Notes
(c) (ii) (iii)	(ii) $1 \times 3^3 \times 2 = 54$ (iii) $3 \times 2 \times 2 \times 2 \times 2 = 48$	Scale 10D (0, 3, 5, 7, 10) Accept correct answer(s) without work <i>Low Partial Credit:</i> • Work of merit in one part, for example, lists some correct patterns <i>Mid Partial Credit</i> • One part correct • Work of merit in both parts <i>High Partial Credit</i> • One part correct and work of merit in the other part

